



CS752 - Mini Project

DFARM: A Deadline-Aware Fault-Tolerant Scheduler for Cloud Computing

Implementation in Edge Layer

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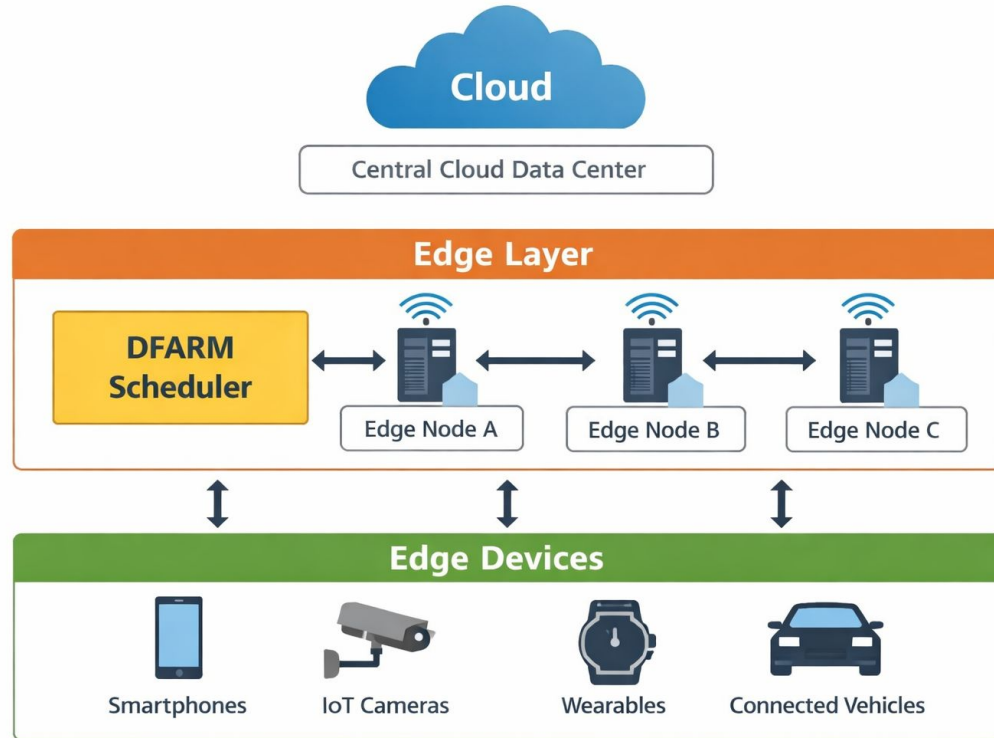
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Problem Statement & Architecture

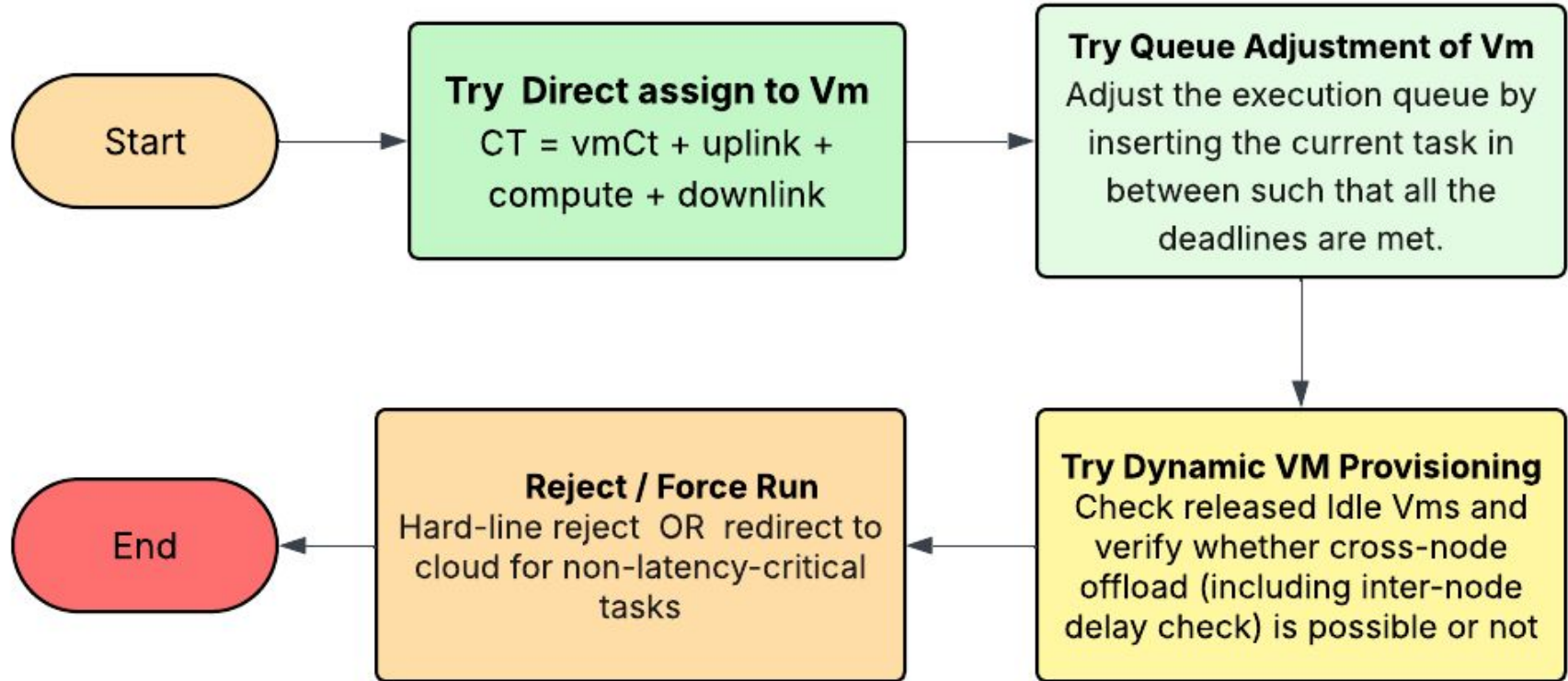
DFARM (Deadline and Fault-Aware Resource Manager)

is an edge computing scheduler designed to efficiently allocate tasks to distributed edge nodes by considering:

- Task Deadlines
- CrossNode Offloading
- Replication
- Network Delays



DFARM: Scheduling Logic



Experimental Setup

Simulator: CloudSim

Programming Language: Java

Cloudlet Types:

Category	Million Instructions
Small	1000MI
Medium	2000MI
Large	5000MI
Huge	8000MI

Edge Nodes and VMs:

Node Name	No of VMs	MIPS	Boot time`	Acquisition Delay
Node A	4	100,200,300,500	5s	10s
Node B	6	500,750, 900, 1000,1100, 1200	3s	8s
Node C	4	100,250,500,750	8s	15s

Design Overview(Classes and Responsibilities)

EdgeCloudlet

Represents an edge task submitted by an IoT/mobile device, carrying deadline, payload size, and latency constraints

EdgeVm

Represents a virtual machine on an edge node, encapsulating compute power (MIPS), boot/acquisition delays, and bandwidth

EdgeNode

Represents a physical edge infrastructure node, managing VM lifecycle pools and supporting cross-node offloading

DFARMScheduler

Implements the core DFARM scheduling logic, assigning tasks to VMs, queue adjustment, VM acquisition.

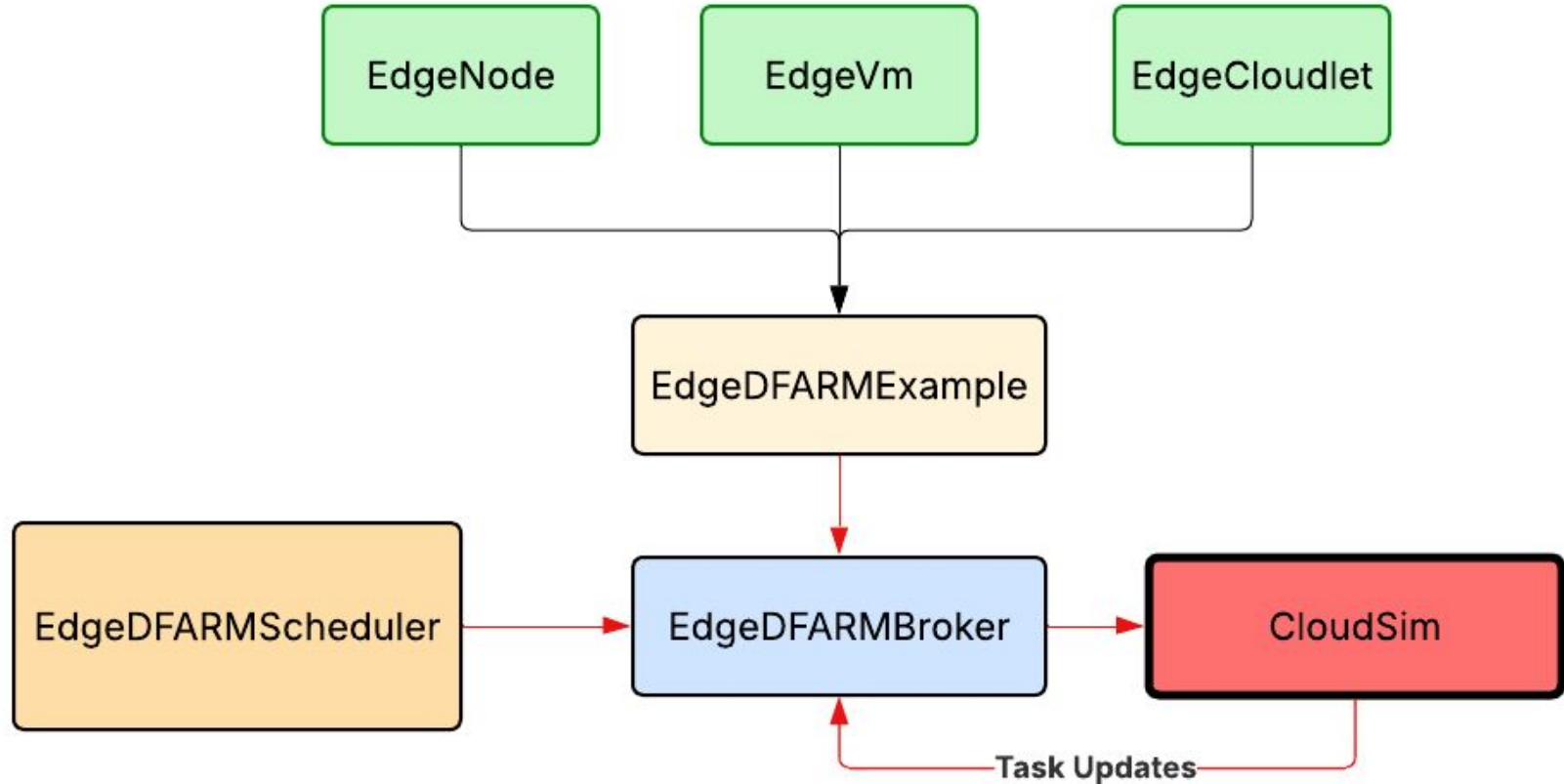
EdgeDFARMBroker

Connects the DFARM scheduler with the CloudSim event loop, handling VM creation, cloudlet submission, and reporting performance metrics.

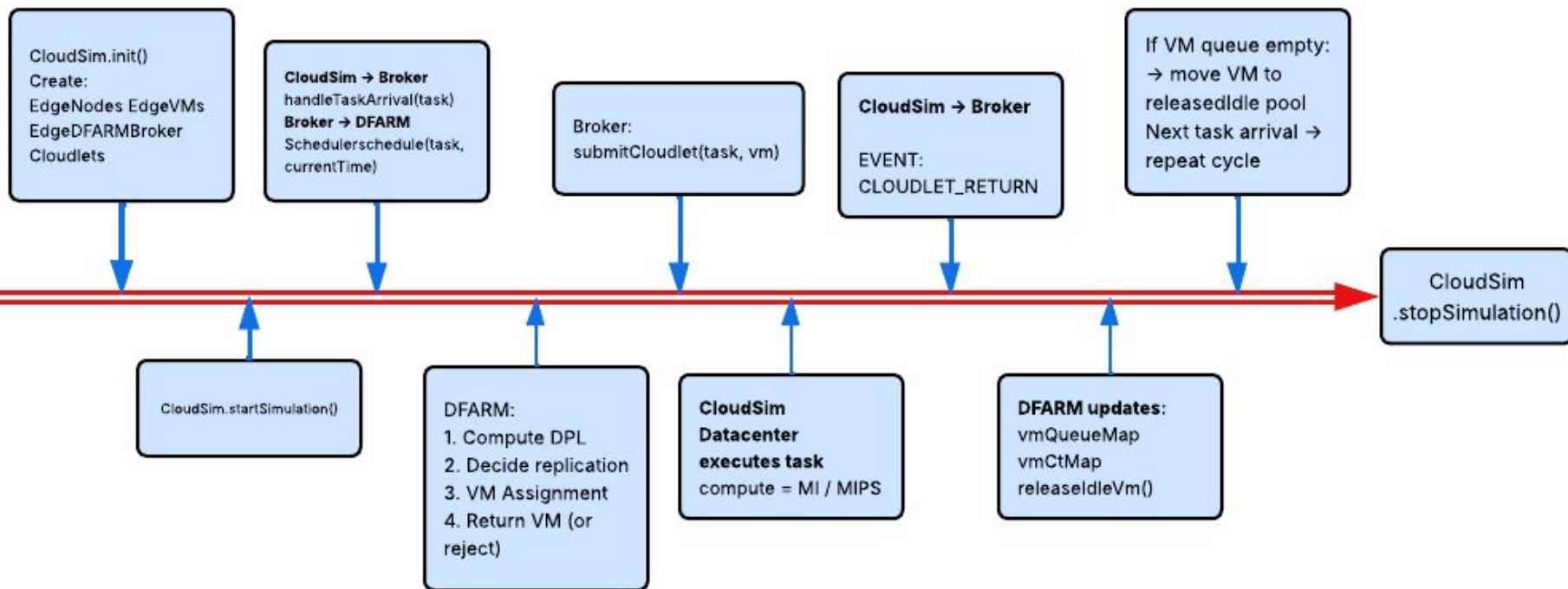
EdgeDFARMExample

Serves as the main simulation driver, configuring edge topology, generating the task workload, launching the simulation.

Design Overview(Classes and Responsibilities)



CloudSim : Simulation Timeline



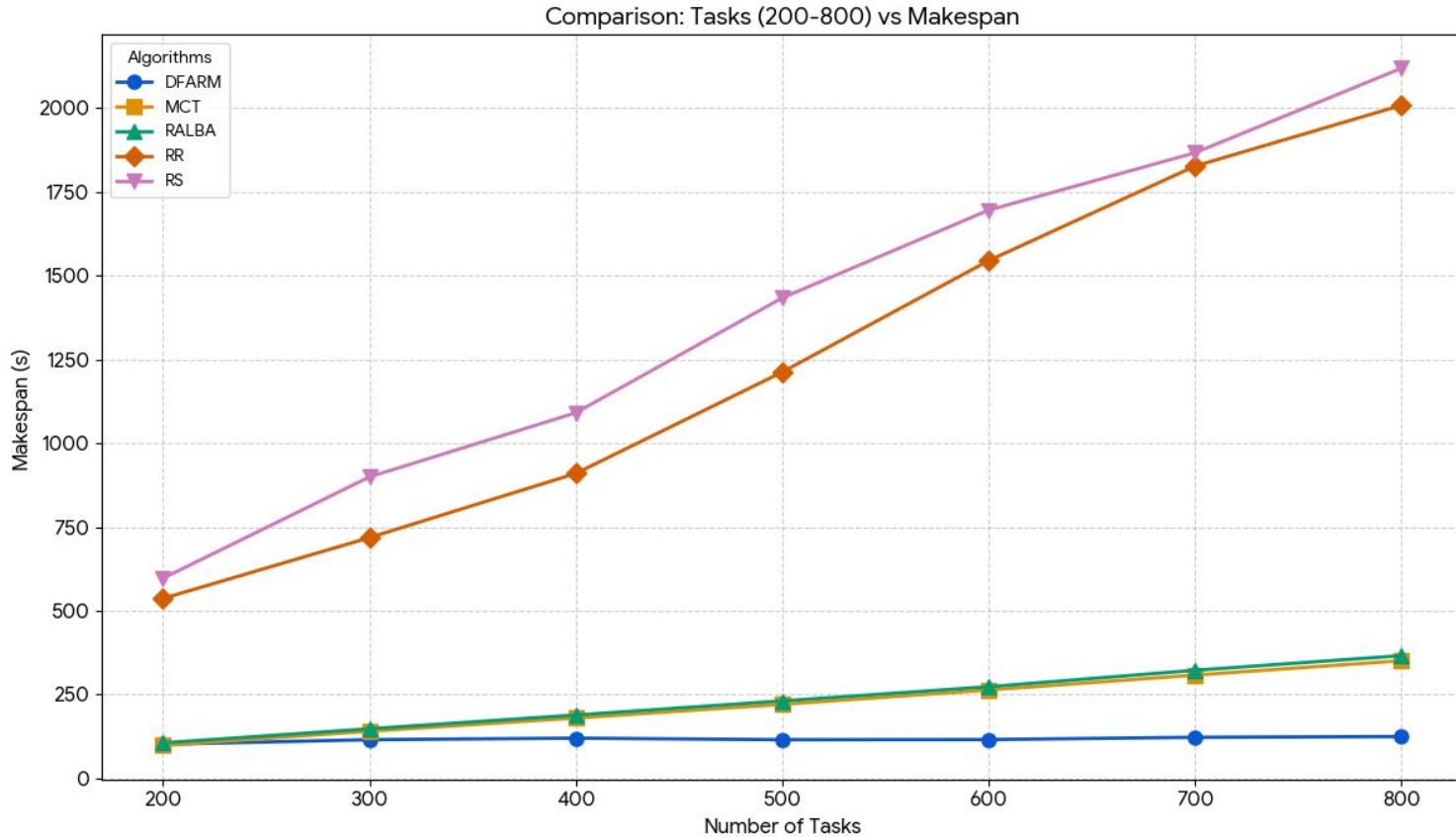
Baseline algorithms for comparison

1. Random Selection (RS): Assign each incoming task to a randomly selected VM.
2. Round Robin (RR): Assign tasks to VMs in cyclic order.
3. Minimum Completion Time (MCT): Assign each task to the VM where it will finish earliest.
4. RALBA (Resource-Aware Load Balancing Algorithm): Distribute tasks evenly using a Fill + Spill strategy.

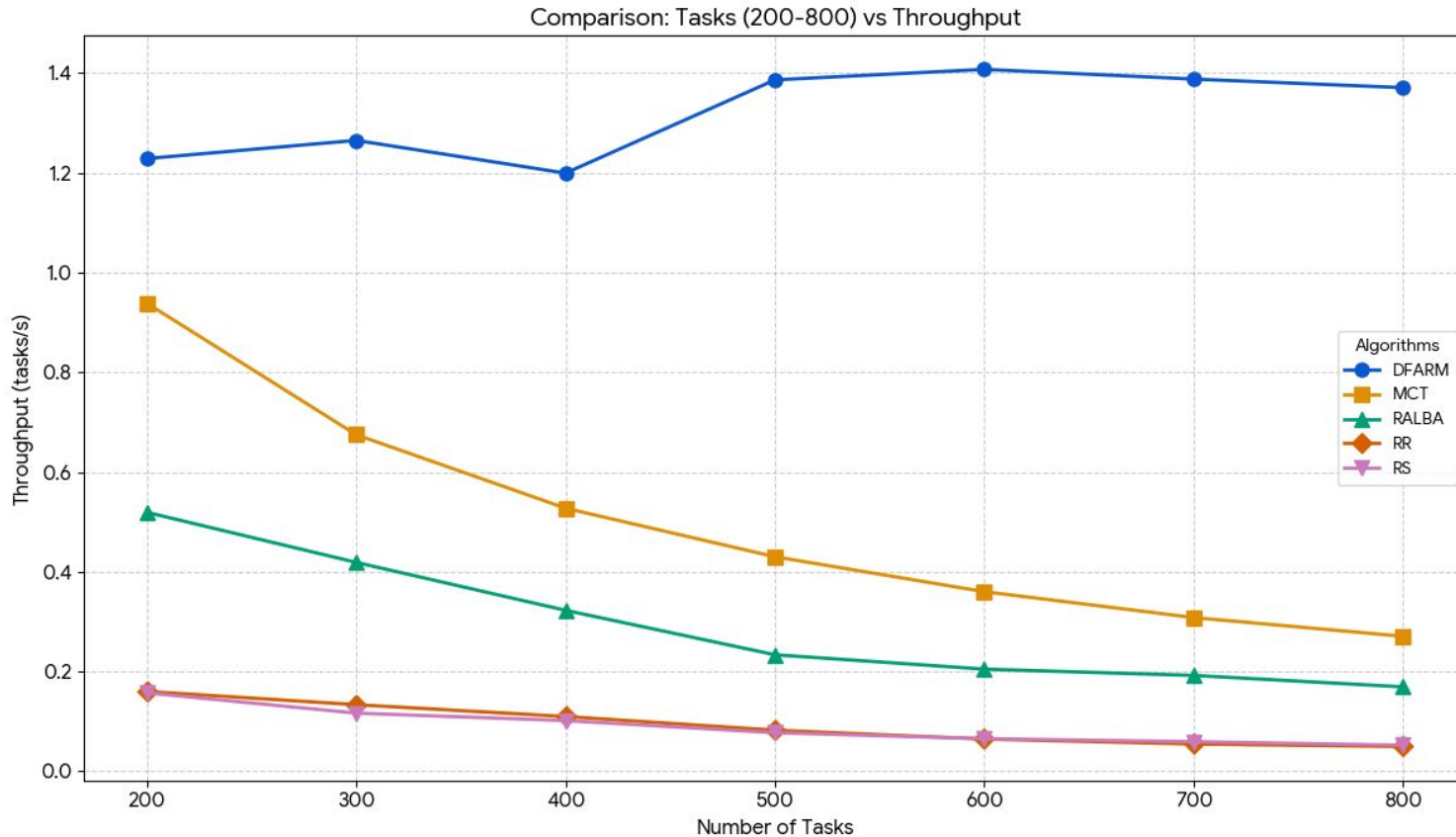
Fill phase: Assign largest tasks to VMs available capacity is filled.

Spill phase: Remaining tasks go to VMs with earliest completion time.

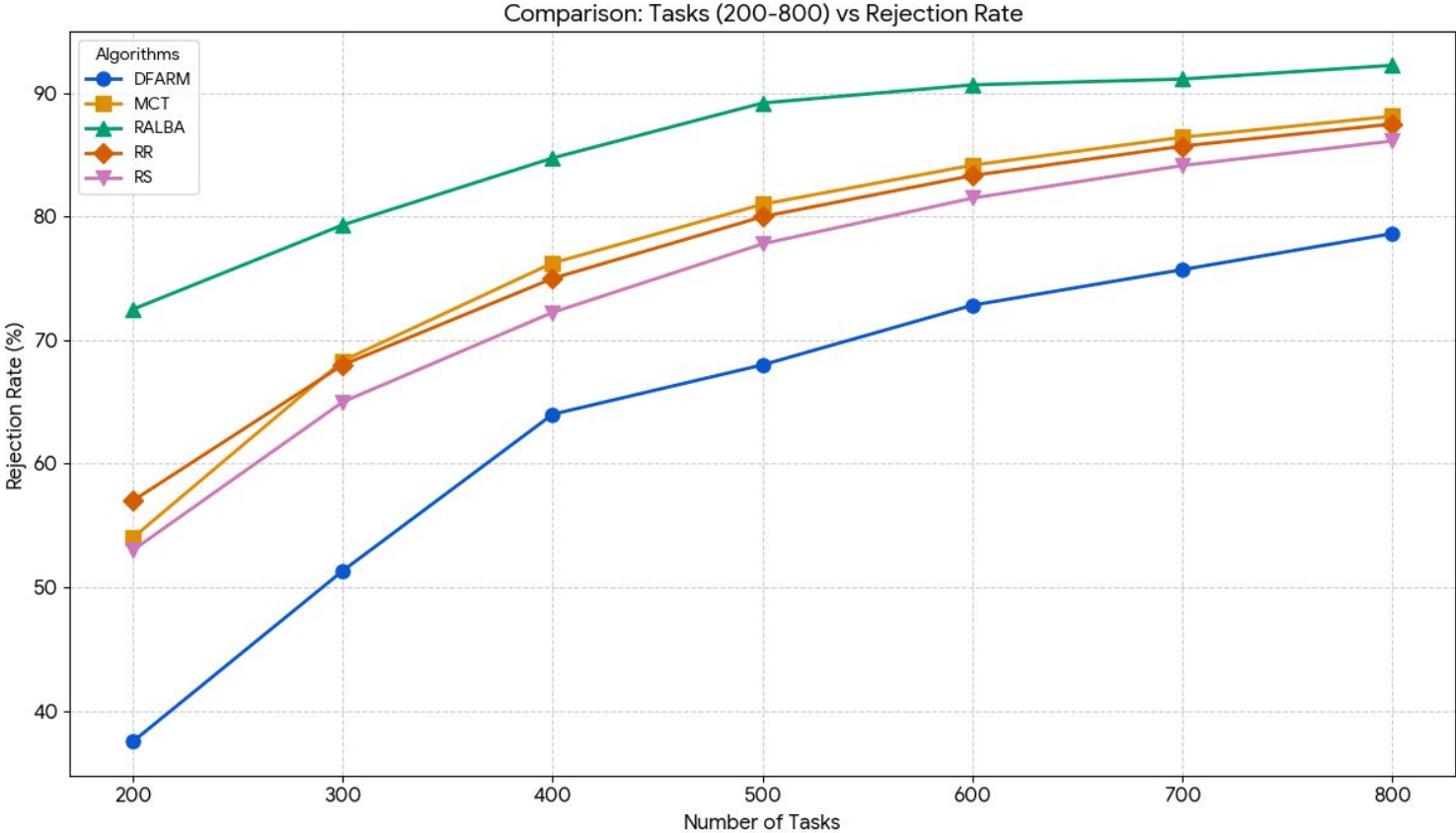
DFARM Results : MakeSpan



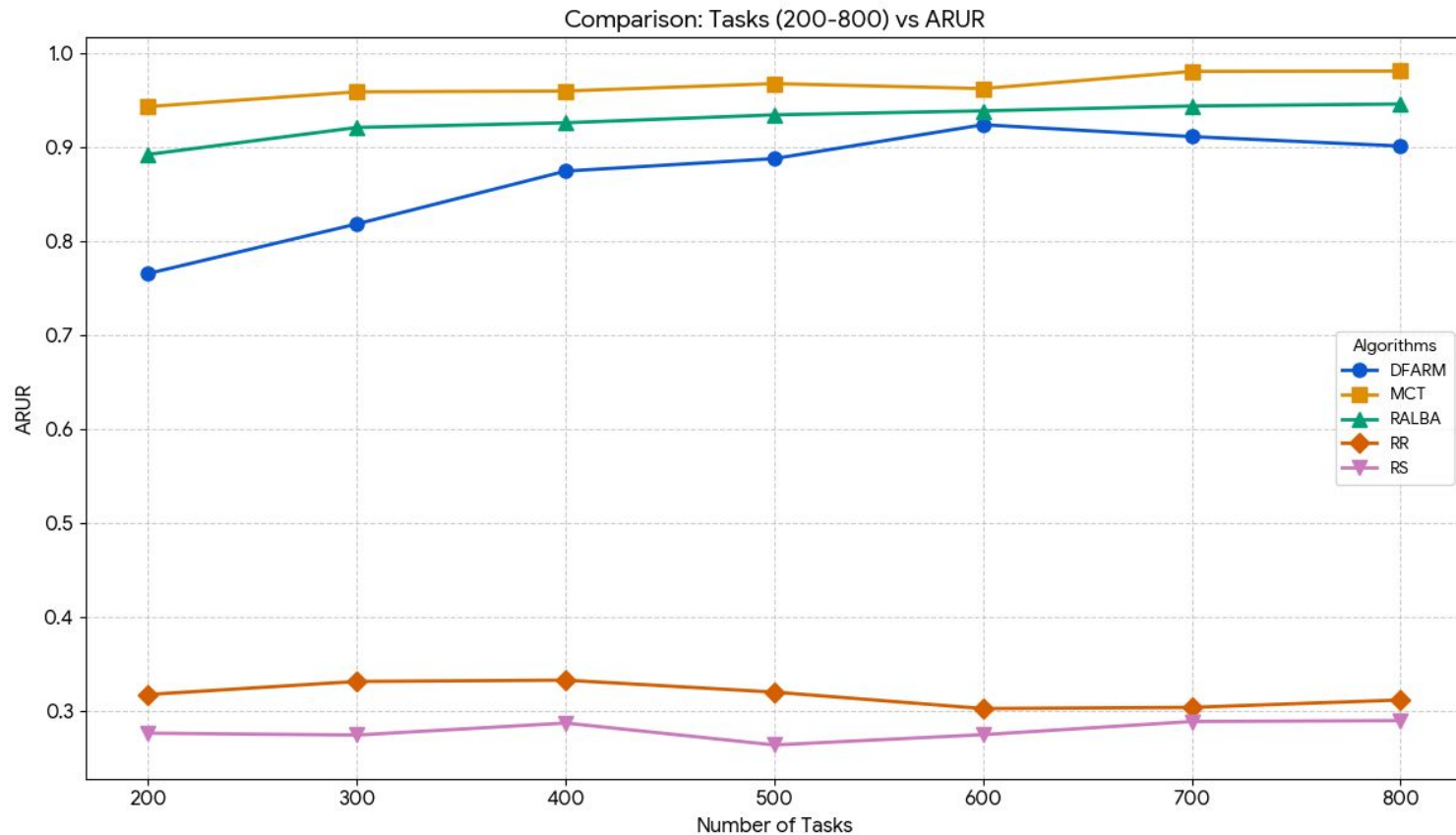
DFARM Results : Throughput



DFARM Results : Rejection Rate(%)



DFARM Results : Average Resource Utilization Ratio



DFARM Results : Summary

Throughput: DFARM maintains a consistently high and stable throughput (averaging ~1.3 tasks/s), significantly outperforming RS and RR, which struggle at levels below 0.2 tasks/s.

Makespan: DFARM achieves the most efficient execution time, remaining stable near 120s even as workload increases, while RS and RR escalate sharply to over 2,000s.

Rejection Rate: DFARM consistently records the lowest rejection rate among all algorithms, demonstrating its superior ability to meet task deadlines under heavy load.

ARUR: DFARM exhibits high resource efficiency with utilization climbing toward 90%, Replication is one of the reasons that makes DFARM slightly performing under MCT and RALBA.

References

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THANK YOU
